Lab 3 – Osmosis in potato strips

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Osmosis = net movement of WATER across a partially permeable membrane, from high water potential to low.

Method: potato strips soaked 30 min in sucrose solutions, mass measured before and after.

Results (% change in mass):

- 0.0 M sucrose -> +9 %
- 0.4 M sucrose -> 0 % (no net change)
- 0.8 M sucrose -> -14 % (strips limp, plasmolysed)

Conclusion: potato cells are isotonic with about 0.4 M sucrose. In 0.8 M the cells lost water.

Key words: hypertonic, hypotonic, isotonic, turgor, plasmolysis, water potential